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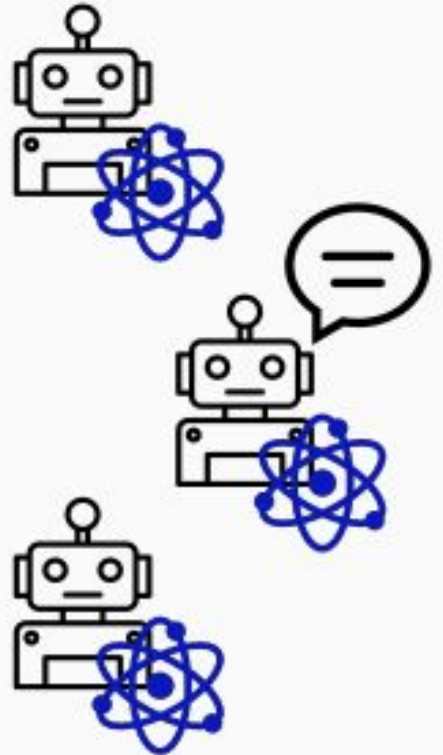
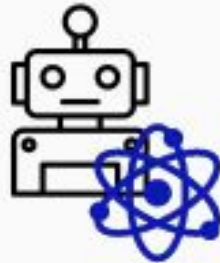
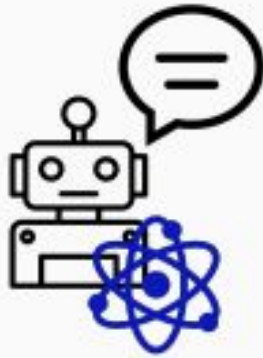


Quantum Protocol Verification

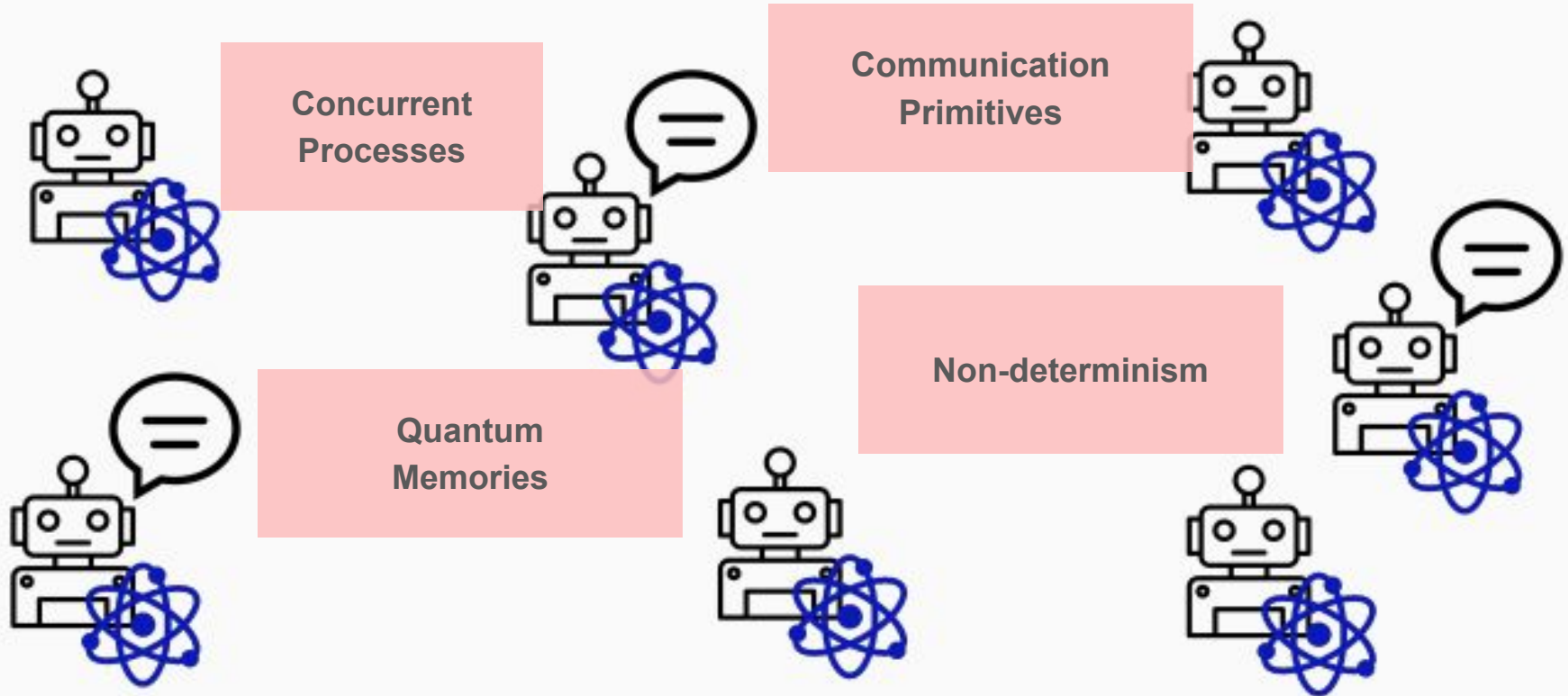
Lorenzo Ceragioli @ IMT Lucca

Joint Work with Fabio Gadducci, Giuseppe Lomurno @ UniPI
and Gabriele Tedeschi @ IRIF

Communicating Quantum Capable Agents



Communicating Quantum Capable Agents



Verification Framework

- **Description Language:**
process calculi (lqCCS)
- **Operational Semantics:** probabilistic LTS, pLTS, ...
- **Behavioural Equivalence:** Bisimilarity, Trace or Testing Equivalence
Correctness = $P \sim \text{Spec}$
Security = $P1 \sim P2$
 $P \parallel \text{Eve} \sim \text{Bad}$

Linear Quantum CCS

$$\langle |\psi\rangle, P \rangle \in \text{Conf}$$

$$P ::= \mathbf{0}_{\tilde{q}} \mid \tau.P \mid c!v.P \mid c?x.P$$

$$\mid P + P \mid P \parallel P \mid \text{if } e \text{ then } P \text{ else } P \mid P \setminus c$$

$$\mid U(\tilde{e}).P \mid M_B(\tilde{e} \triangleright x).P$$

$$\langle \text{Conf}, \text{Act}, \rightarrow \subseteq \text{Conf} \times \text{Act} \times \mathcal{D}(\text{Conf}) \rangle$$

$$\langle |\psi\rangle, M_{\{b_0, b_1\}}(\tilde{q} \triangleright y).P \rangle \xrightarrow{\tau} \langle |\phi_0\rangle, P[0/y] \rangle_{M_0^{\tilde{q}}|\psi} \oplus \langle |\phi_1\rangle, P[1/y] \rangle$$

$$\langle |00\rangle, H(q_1).M_{01}(q_1 \triangleright y).c!y.\mathbf{0}_{q_1} \rangle$$



$$\langle | +0 \rangle, M_{01}(q_1 \triangleright y).c!y.\mathbf{0}_{q_1} \rangle$$



$$\langle |00\rangle, c!0.\mathbf{0}_{q_1} \rangle$$

$$\langle |10\rangle, c!1.\mathbf{0}_{q_1} \rangle$$



$$\langle |00\rangle, \mathbf{0}_{q_1} \rangle$$



$$\langle |10\rangle, \mathbf{0}_{q_1} \rangle$$

Value Passing Semantics

$$\frac{}{c!v.P \xrightarrow{c!v} P}$$

$$\frac{}{c?x.P \xrightarrow{c?v} P[v/x]}$$

$$\frac{P \xrightarrow{c!v} P' \quad Q \xrightarrow{c?v} Q'}{P \parallel Q \xrightarrow{\tau} P' \parallel Q'}$$

$$\langle | + 0 \rangle, c!q_1.\mathbf{0}_{q_2} \rangle \xrightarrow{c!q_1} \bigcirc \rightsquigarrow \langle | + 0 \rangle, \mathbf{0}_{q_2} \rangle$$

Matching Physical Reality – Value Passing

$$\frac{}{c!v.P \xrightarrow{c!v} P}$$

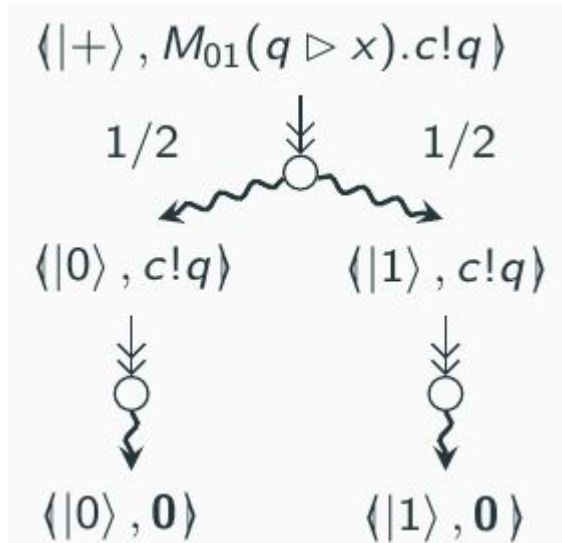
$$\frac{}{c?x.P \xrightarrow{c?v} P[v/x]}$$

$$\frac{P \xrightarrow{c!v} P' \quad Q \xrightarrow{c?v} Q'}{P \parallel Q \xrightarrow{\tau} P' \parallel Q'}$$

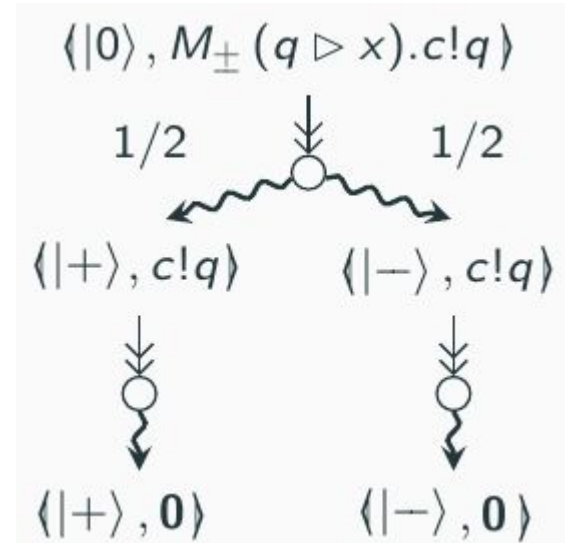
$$\langle |+\mathbf{0}\rangle, c!q_1.\mathbf{0}_{q_2} \rangle \xrightarrow{c!q_1} \langle |+\mathbf{0}\rangle, \mathbf{0}_{q_2} \rangle$$

“Hidden” private state is revealed

Matching Physical Reality – Distributions



VS



Matching Physical Reality – Non-determinism

Every choice is legal

But not their combination

Unconstrained non-deterministic
processes nullify distributions

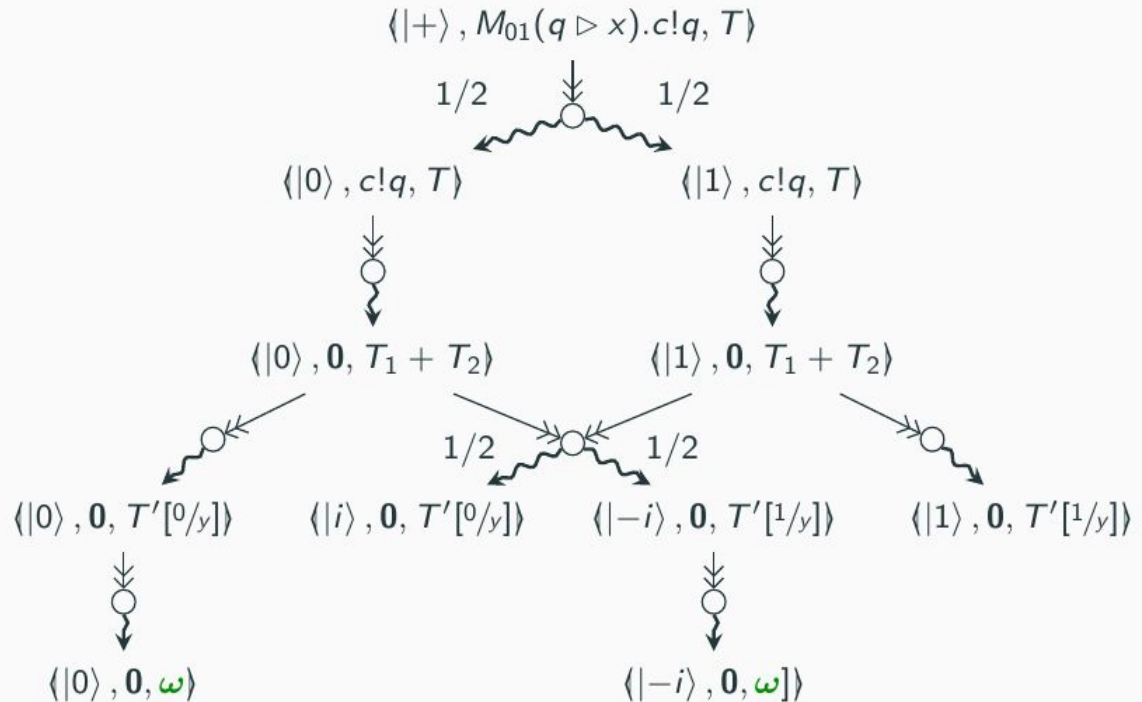
Problems:

Trace equivalence

Bisimilarity

Testing Equivalence

Example: quantum coin toss



Matching Physical Reality – Non-determinism

Every choice is legal

But not their combination

Unconstrained non-deterministic
processes nullify distributions

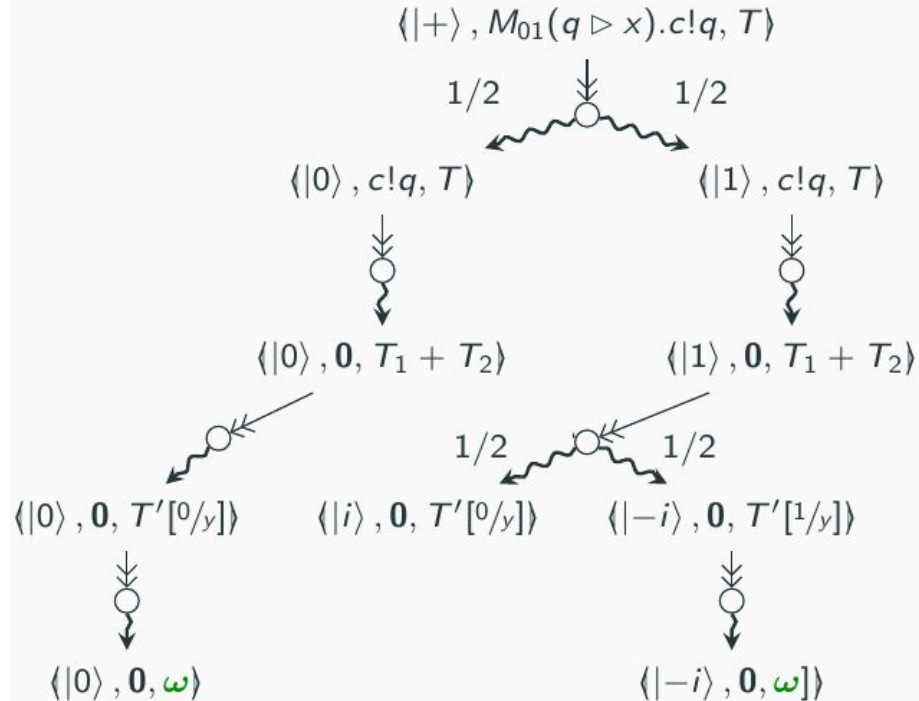
Problems:

Trace equivalence

Bisimilarity

Testing Equivalence

Example: **quantum coin toss**



Solution via Physically Admissible Schedulers

- Label each choice in processes
- Impose consistent scheduler for each move based on classical values only

Theorem:

From indistinguishable distributions of quantum states
to indistinguishable distributions of lqCCS configurations

Different Bisimilarities

Table 1. Recap of the main differences between the proposed bisimilarities.

PAIR OF PROCESSES (in lqCCS syntax)	QPAIg	CQP	qCCS	lqCCS
$c?x.H(x).0$ and $c?x.X(x).0$	$\sim$	$\sim$	$\not\sim_p, \not\sim_d$	illegal
$c?x.H(x).0_x$ and $c?x.X(x).0_x$	$\sim$	$\sim$	$\sim_p, \sim_d$	$\sim_s, \sim_{cs}$
$c?x.H(x).d!x$ and $c?x.X(x).d!x$	$\not\sim$	$\not\sim$	$\not\sim_p, \not\sim_d$	$\not\sim_s, \not\sim_{cs}$
$\text{Set}_{\frac{1}{4}I}(q_1, q_2).(c!q_1 \parallel c!q_2)$ and $\text{Set}_{ \Phi^+ \langle\Phi^+ }(q_1, q_2).(c!q_1 \parallel c!q_2)$	$\sim$	$\not\sim$	$\not\sim_p, \not\sim_d$	$\not\sim_s, \not\sim_{cs}$
$\text{Set}_{ +\chi+ }(q).M_{01}(q \triangleright x).c!q$ and $\text{Set}_{ 0\chi 0 }(q).M_{\pm}(q \triangleright x).c!q$	$\not\sim$	$\sim$	$\not\sim_p, \sim_d$	$\not\sim_s, \sim_{cs}$
$\text{Set}_{ +\chi+ }(q).M_{01}(q \triangleright x).(c!q + d!q)$ and $\text{Set}_{ 0\chi 0 }(q).M_{\pm}(q \triangleright x).(c!q + d!q)$	$\not\sim$	$\sim$	$\not\sim_p, \sim_d$	$\not\sim_s, \not\sim_{cs}$

Decidability Problem: Infinite Input States

$$\forall |\psi\rangle . \langle |\psi\rangle , P \rangle \sim \langle |\psi\rangle , Q \rangle$$

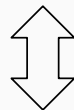
Approach

From parametric pLTS

To input agnostic QLTS

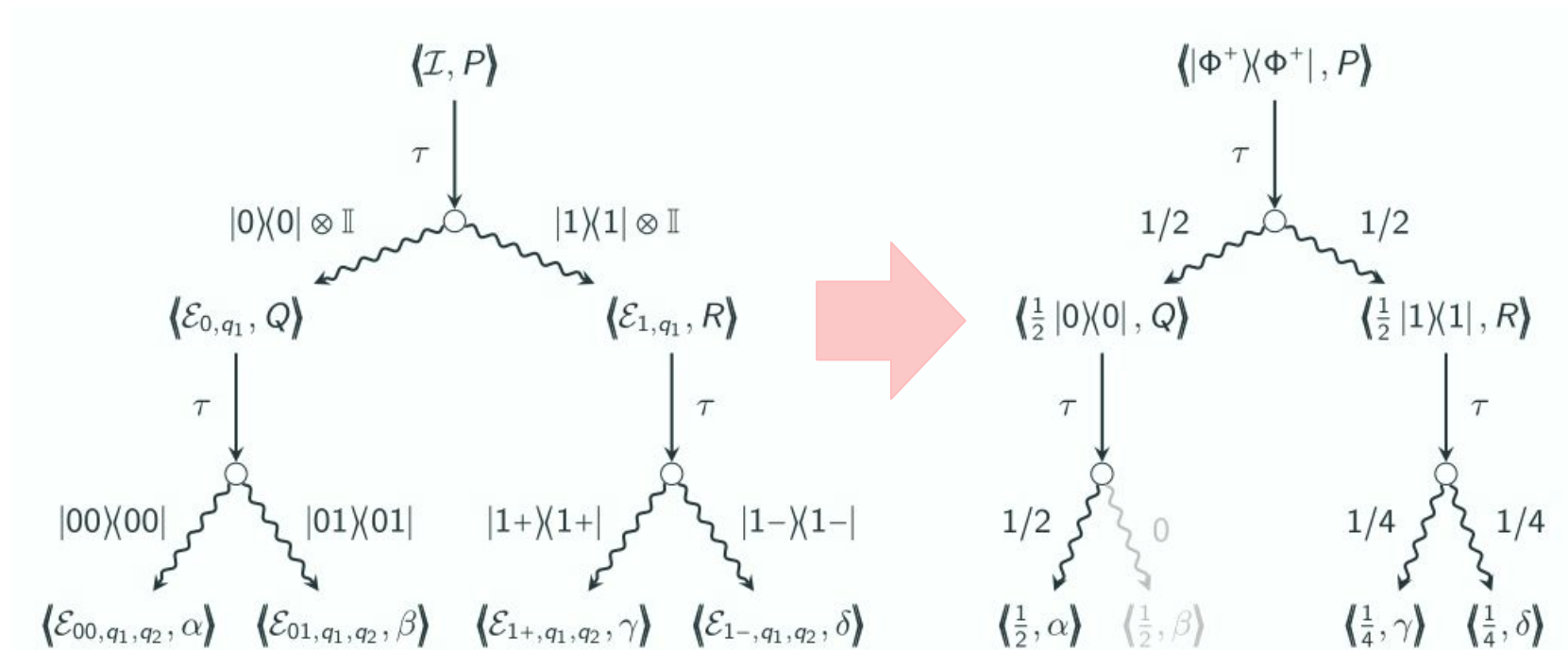
Theorem

$$\forall |\psi\rangle . \langle |\psi\rangle , P \rangle \sim \langle |\psi\rangle , Q \rangle$$

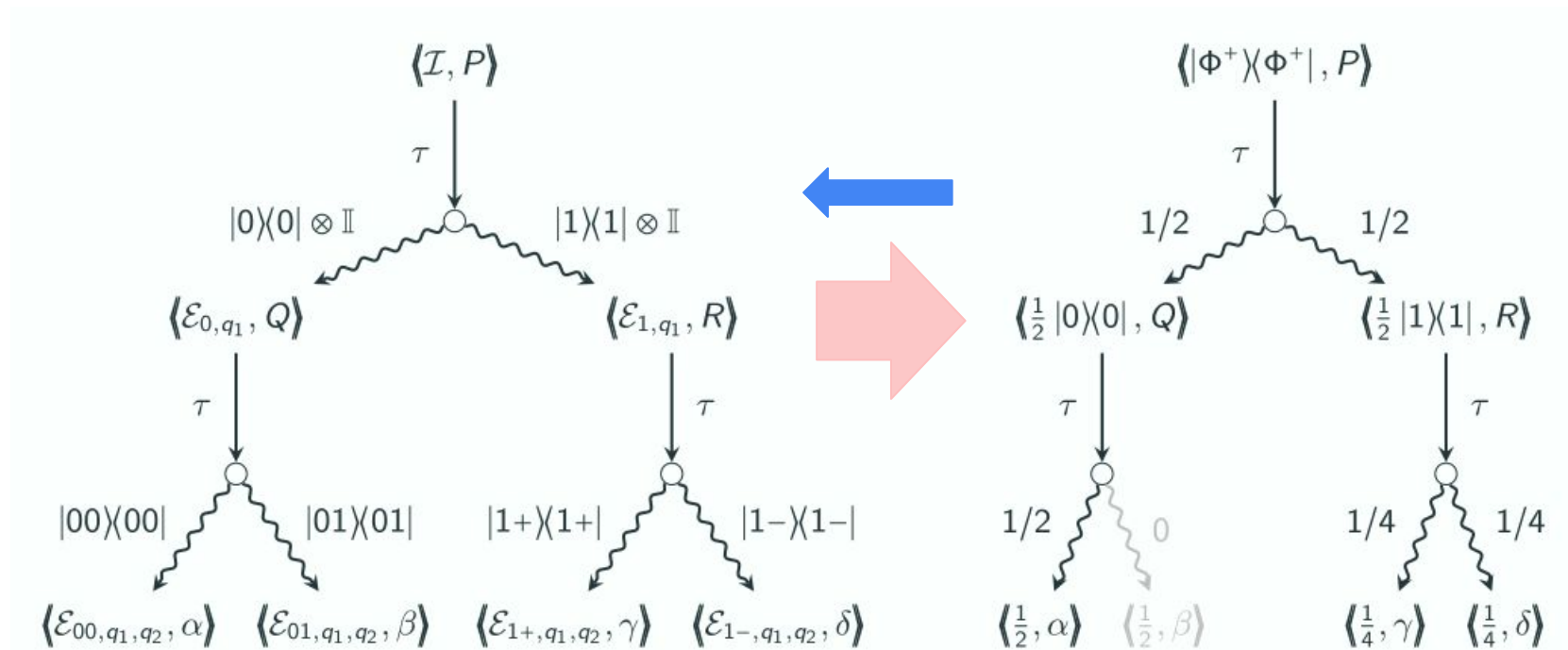


$$\langle \mathcal{I}, P \rangle \simeq \langle \mathcal{I}, Q \rangle$$

Decidability Problem: Infinite Input States



Decidability Problem: Infinite Input States



Future Work

- Characterize the Strongest Good Schedulers
- Additional Classical Features
- General Result on Decidability and Congruence
- Verification Tool
- More Real-world Protocols

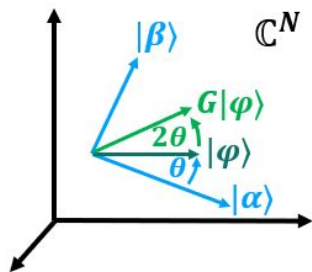
Publications

- L. Ceragioli, F. Gadducci, G. Lomurno, G. Tedeschi: **Quantum Bisimilarity via Barbs and Contexts: Curbing the Power of Non-deterministic Observers**. POPL 2024
- L. Ceragioli, G. Lomurno, G. Tedeschi: **Reconciling Quantum Theory and Process Equivalence via Physically Admissible Schedulers**. WADT 2024
- L. Ceragioli, F. Gadducci, G. Lomurno, G. Tedeschi: **Quantum Bisimilarity Is a Congruence Under Physically Admissible Schedulers**. APLAS 2024
- L. Ceragioli, F. Gadducci, G. Lomurno, G. Tedeschi: **Effect Semantics for Quantum Process Calculi**. CONCUR 2024
- L. Ceragioli, F. Gadducci, G. Lomurno, G. Tedeschi: **Testing Quantum Processes**. ISoLA 2024
- L. Ceragioli, E. Di Lavore, G. Lomurno, G. Tedeschi: **A Coalgebraic Model of Quantum Bisimulation**. ACT 2024

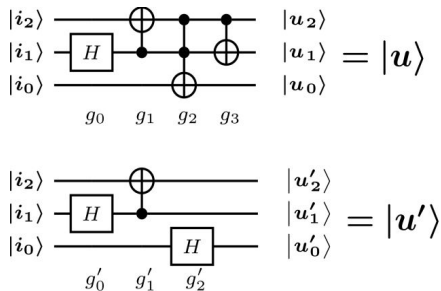
Extra: Automated Quantum Stack



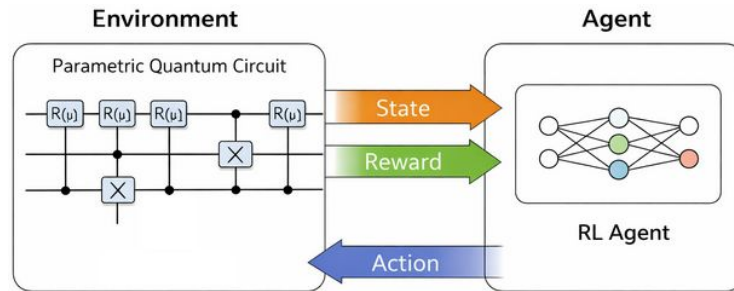
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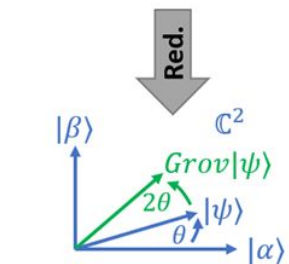
Simulation



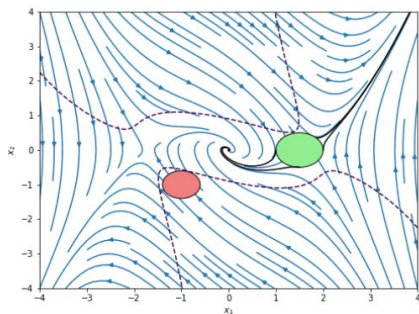
Verification



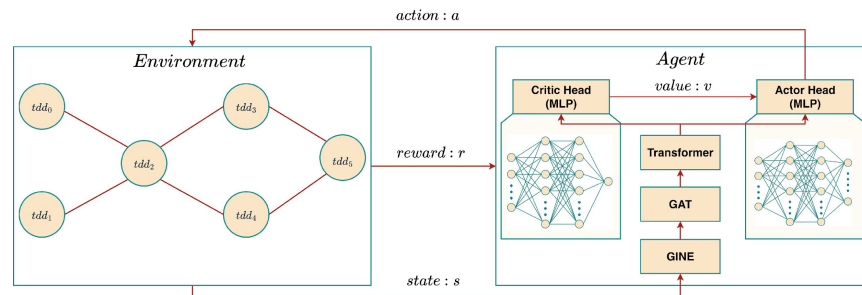
Synthesis



via Abstraction

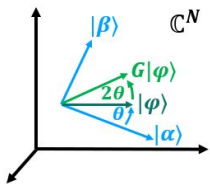


via Lyapunov Theory

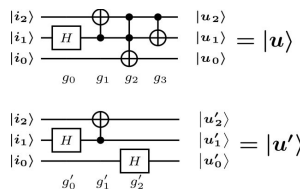


via Reinforcement Learning

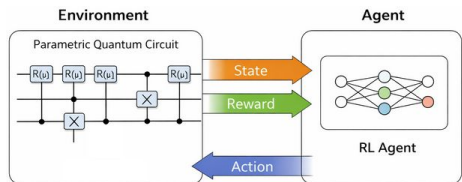
Automated Quantum Stack



Simulation



Verification



Synthesis

Papers and Funding:

POUL DUE JENSEN GRUNDFOS
FOUNDATION

DeiC
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**INDEPENDENT
RESEARCH FUND
DENMARK**

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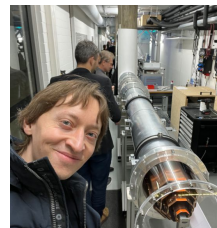


L. Burgholzer, A. Jiménez-Pastor, K. G. Larsen, M. Tribastone, M. Tschaikowski, R. Wille.
Forward and Backward Constrained Bisimulations for Quantum Circuits Using Decision
Diagrams. **ACM Trans. Quantum Comput.** 2025

E. Casablanca, O. Schön, P. Zuliani, S. Soudjani. LUCID: Learning-Enabled Uncertainty-Aware
Certification of Stochastic Dynamical Systems. **AAAI 2026**



Mirco Tribastone



Max Tschaikowski



Andrea Vandin



Paolo Zuliani



Extra from Pisa – QCOMICAL

QCOMICAL is an international research network on *quantum and classical programming languages and semantics*

- Fabio Gadducci is the coordinator for Pisa

Italia:

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Università degli Studi di Urbino Carlo Bo

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+ Argentina, Francia, Uruguay

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12 al 16 novembre al Museo
degli Strumenti per il Calcolo
~ 60 partecipanti nel 2025